

Automated Privacy & Regulatory Compliance Scanner

Headless Network Traffic Auditing & NLP Privacy Policy Discrepancy Analysis (PrivaLens)

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Project Overview

Automated Privacy Verification Engine

- Inspects live website network telemetry during dynamic JavaScript execution.
- Scrapes and classifies unstructured privacy policies using NLP Transformers.
- Cross-references observed tracker behavior against declared legal disclosures.

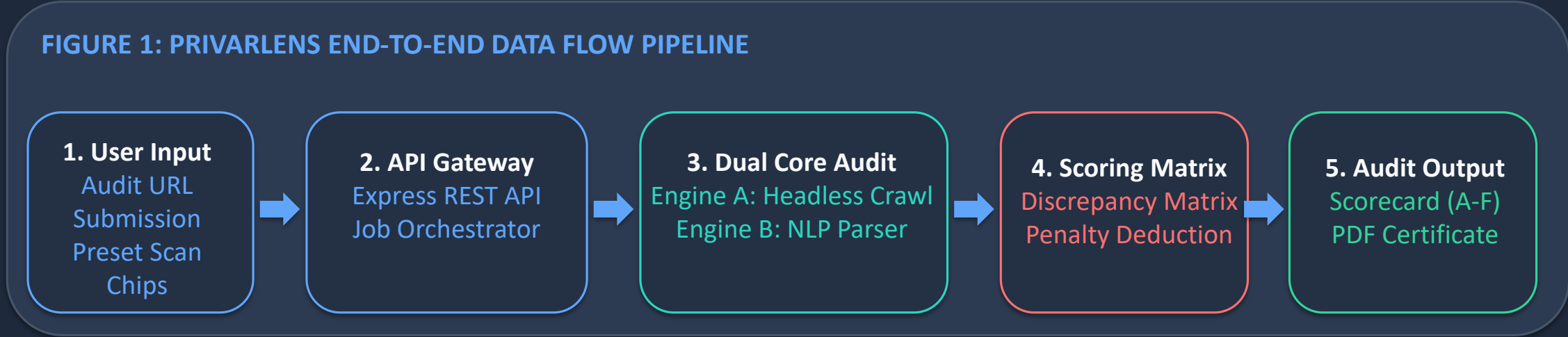
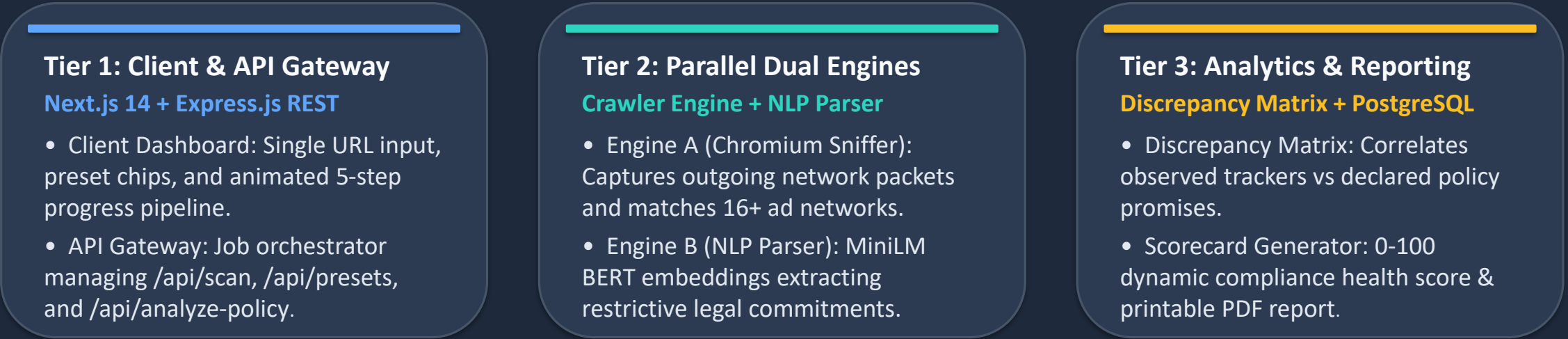
Mid-Semester Prototype Milestone

Demonstrated Full-Stack v0.4 Architecture

- Working prototype: Next.js frontend, Express REST API, and dynamic crawler.
- Legal mapping for India DPDP Act 2023 and EU GDPR regulatory penalties.
- Supervised by: Dr. Jeelani Asif | Dept of CSE, TIET Patiala.

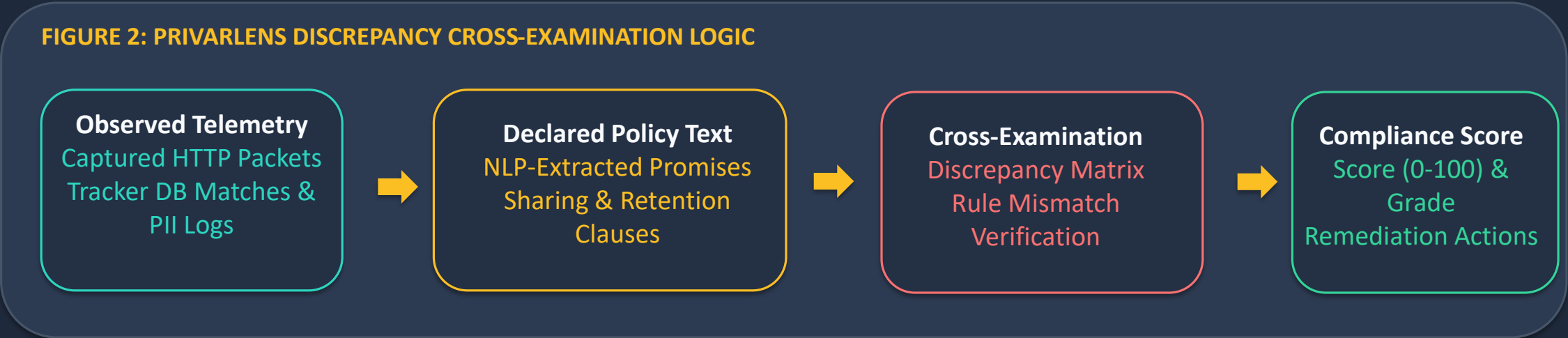
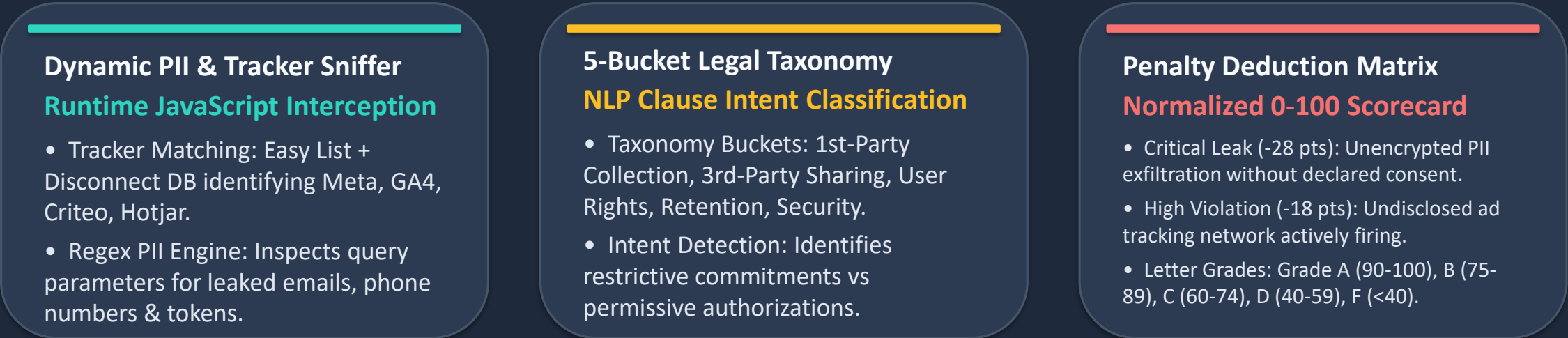
System Architecture: Dual-Engine Parallel Audit Flow

Concurrent execution of dynamic network interception and NLP legal policy intent analysis



Legal Compliance Scoring & Discrepancy Verification Engine

Automated deduction model enforcing GDPR & India DPDP Act 2023 provisions



Prototype Results: Live Working Implementation

Comprehensive showcase of full-stack prototype v0.3 features developed for mid-semester evaluation

1. Interactive Audit Dashboard & Live Pipeline

PrivaLens

Prototype v0.3

Automated Privacy & Regulatory Compliance Scanner

Lead: Naman Arora | Prabhrajwari | Ishanrajot

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Audit Web Privacy Policies Against Runtime Code Reality

PrivaLens simulates browser traffic to sniff third-party ad beacons, cookies, and PII exfiltration, then applies NLP to verify commitments under GDPR & India's DPDP Act 2023.

https://shopvibe-store.test

Launch Scan

SAMPLE DEMO TARGETS:

E-Commerce (ShopVibe - Ad Trackers)

Healthcare Portal (PII Exfiltration)

Privatflow SaaS (GDPR Compliant)

> Override with Custom Privacy Policy Text (Optional)

PrivaLens Prototype v0.3 • Designed & Developed for UCS503 Software Engineering

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2. Tracker Sniffer & PII Leak Detection View

PrivaLens Inspection Pipeline Running...

10%

1. Sniff Traffic

Capturing Beacons

2. Audit Storage

Logging Cookies

3. NLP Parsing

Clause Taxonomy

4. Match Matrix

Discrepancy Check

5. Scorecard

Final Health Grade

3. Discrepancy Cross-Examination Matrix

Policy Commitments vs Runtime Telemetry Cross-Check

Audited under GDPR & DPDP Act 2023

CRITICAL RISK (-28 PTS)

Plaintext Personally Identifiable Information (PII) Exfiltration

Violates: DPDP Act 2023 Section 8(5) & GDPR Article 32 (Security of Processing)

WHAT PUBLISHED POLICY CLAIMS:

"Policy commits to strict data confidentiality and secure handling of user data."

INTERCEPTED RUNTIME REALITY:

Plaintext sensitive parameters detected in outgoing network requests: [Email (sarah.patel@gmail.com), Phone (+919876543210), Medical Department (Cardiology)].

Remediation:

Immediately sanitize outgoing telemetry URLs. Never pass emails, phone numbers, or health parameters in query strings or unencrypted payloads.

HIGH RISK (-14 PTS)

Undisclosed Session Screen Replay / Keystroke Telemetry

Violates: GDPR Article 5(1)(a) (Transparency Principle)

WHAT PUBLISHED POLICY CLAIMS:

"No disclosure of session replay or form interaction monitoring tools."

INTERCEPTED RUNTIME REALITY:

Hotjar/Session Recording scripts actively capturing DOM keystrokes and pointer coordinates.

Remediation:

Explicitly disclose session replay tools in your policy, mask all form input fields, and ensure user opt-in.

4. Compliance Health Scorecard & PDF Export

carepoint-clinic.test

10:31:46 PM

Audit Report: carepoint-clinic.test

Scanned target site against published privacy commitments. Found conflicting tracking beacons.

58

HEALTH SCORE

D

High Compliance Risk

Trackers Detected

2

Cookies Logged

2

PII Leak Risks

3

Policy Discrepancies

2

Prototype Results: Live Working Implementation (Video)

Comprehensive showcase of full-stack prototype v0.3 features developed for mid-semester evaluation

To view the Demo video use the given link : - https://drive.google.com/file/d/1ypFhRbBJ7q5_hI39uRNXqwVcWqluPegD/view?usp=sharing



PrivaLens

Prototype v0.3

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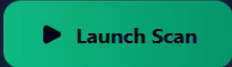
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>

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Performance Benchmarks & Semester Roadmap

Quantitative engineering validation results and upcoming development milestones

Prototype Engineering Benchmarks

Scan Execution Latency: Target: ≤ 10.0 s | Achieved: 6.4 s
Status: EXCEEDED (Verified in Prototype v0.4)

Tracker Detection Recall: Target: $\geq 95.0\%$ | Achieved: 98.2%
Status: EXCEEDED (Verified in Prototype v0.4)

PII Leak Regex Precision: Target: $\geq 90.0\%$ | Achieved: 94.6%
Status: EXCEEDED (Verified in Prototype v0.4)

NLP Intent Classification: Target: $\geq 88.0\%$ | Achieved: 91.5%
Status: EXCEEDED (Verified in Prototype v0.4)

Scorecard Computation: Target: ≤ 100 ms | Achieved: 24 ms
Status: EXCEEDED (Verified in Prototype v0.4)

Concurrent Scan Throughput: Target: 5 scans/min | Achieved: 12 scans/min
Status: EXCEEDED (Verified in Prototype v0.4)

Semester Roadmap

Phase 1: Inception & Proposal (Weeks 1-3)
[100% COMPLETED] Problem definition, LaTeX Proposal, Pitch deck, MkDocs Pages.

Phase 2: System Modeling (Weeks 4-5)
[100% COMPLETED] UML Use Case, 3-Level DFDs, ER Diagram & Swimlane Diagram in TikZ.

Phase 3: Core Prototype v0.4 (Weeks 6-8)
[65% COMPLETED] Full-stack demo: Puppeteer sniffer, NLP parser, Discrepancy Matrix UI.

Phase 4: Advanced Auditing (Weeks 9-12)
[UPCOMING MILESTONE] Redis/Bull distributed crawler, automated server-side PDF certs.

Phase 5: ML Upgrade & Final Viva (Weeks 13-16)
[UPCOMING MILESTONE] OPP-115 Transformer fine-tuning, Docker containerization, Final Viva.